**Experiment 6**

**Construct a C program to simulate Round Robin scheduling algorithmwith C.**

```
#include <stdio.h>

int main()
{
   int n, i, tq, time = 0, completed = 0;
   int bt[20], rt[20], wt[20], tat[20], at[20];

   float avg_wt = 0, avg_tat = 0;

   printf("Enter number of processes: ");
   scanf("%d", &n);

   for (i = 0; i < n; i++)
   {
      printf("Enter arrival time and burst time of P%d: ", i + 1);
      scanf("%d%d", &at[i], &bt[i]);
      rt[i] = bt[i];
   }

   printf("Enter time quantum: ");
   scanf("%d", &tq);

   while (completed < n)
   {
      int executed = 0;

      for (i = 0; i < n; i++)
      {
         if (rt[i] > 0 && at[i] <= time)
```

```
      {
         executed = 1;

         if (rt[i] > tq)
         {
            time += tq;
            rt[i] -= tq;
         }
         else
         {
            time += rt[i];
            rt[i] = 0;
            tat[i] = time - at[i];
            wt[i] = tat[i] - bt[i];
            completed++;
         }
      }
   }

   if (executed == 0)
      time++;
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++)
{
   printf("P%d\t%d\t%d\t%d\n",
         i + 1, bt[i], wt[i], tat[i]);

   avg_wt += wt[i];
```

```
        avg_tat += tat[i];
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);

    return 0;
}
```

**Output:**

